

AI STUDYBUDDY API

Data Flow Diagram

Data movement through the StudyBuddy service

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TEAM SIZE	5
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Context

Client -> Express API -> MongoDB for users/material records. Uploaded files are temporarily written under uploads/. Gemini receives prompts containing extracted material text and returns generated learning content.

Authentication

Credentials -> User model/password hashing -> access and refresh JWTs. Protected middleware verifies a Bearer access token and attaches decoded user ID/role.

Upload path

Multipart file -> Multer checks extension and 5 MB limit -> controller reads stored file as UTF-8 -> creates Material document -> deletes temporary file.

AI path

Material lookup -> prompt composed from stored content and request options -> Gemini helper -> summary/flashcards/quiz/studyPlan saved to material -> JSON response.

Privacy boundary

Material text is sent to the configured Gemini service. The project should document user disclosure and provider data handling. Do not include real .env secrets in shared files.